

Finite-Dimensional Positive Case for the Aronszajn-Null Game

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Source and Open Question

The source paper is Jakub Duda and Boaz Tsaban, *Games in Banach spaces*, arXiv:math/0601556. For a separable Banach space X , and a non-zero direction $x \in X$, let $\mathcal{A}(x)$ be the Borel sets whose intersection with every line parallel to x has one-dimensional Lebesgue measure zero. A Borel set $A \subset X$ is Aronszajn-null if for every dense sequence (x_n) in X there are $A_n \in \mathcal{A}(x_n)$ such that

$$A \subset \bigcup_n A_n.$$

Duda–Tsaban define the game $G_{\mathcal{A}}$: in inning n , Player I plays a direction x_n so that the resulting sequence is dense, Player II answers with $A_n \in \mathcal{A}(x_n)$, and Player II wins if $A \subset \bigcup_n A_n$. They prove that if Player I has no winning strategy, then A is Aronszajn-null, and state the converse as open:

$$A \in \mathcal{A} \implies \text{Player I has no winning strategy in } G_{\mathcal{A}}.$$

Partial Result

Theorem 1. *Let X be a finite-dimensional real Banach space and let $A \subset X$ be a Borel Aronszajn-null set. Then Player II has a winning strategy in the Aronszajn-null game $G_{\mathcal{A}}$ played on A .*

Proof Intuition

In finite dimensions, Aronszajn-null Borel sets are Lebesgue null. Suppose Player I has already played independent directions spanning a subspace S . The part of A not yet covered can be over-covered by the inverse image of a null Borel set in the quotient X/S . When Player I next plays a direction whose image in X/S is non-zero, Fubini splits that quotient-null set into the part with null sections parallel to the new direction and a smaller quotient-null residual. Pulling the first part back to X is a legal move for Player II. Since a dense sequence in a finite-dimensional space spans the whole space after finitely many independent moves, this process exhausts the residual.

Proof

We use Lebesgue measure on finite-dimensional quotients of X , normalized arbitrarily; nullness is independent of the normalization.

First note that A is Lebesgue null. Indeed, since A is Aronszajn-null, take any dense sequence containing a basis of X . Then A is covered by countably many sets $B_n \in \mathcal{A}(x_n)$. For any non-zero

x , each Borel $B \in \mathcal{A}(x)$ is Lebesgue null by Fubini over the decomposition of X into lines parallel to x . Hence A is null.

We describe a strategy for Player II. More generally, maintain the following state after some innings. Let $S \subset X$ be the span of the independent directions already accepted, let $\pi : X \rightarrow Q = X/S$ be the quotient map, and let $R \subset Q$ be a Borel null set such that the part of A not yet covered by Player II's previous moves is contained in $\pi^{-1}(R)$. Initially $S = \{0\}$, $Q = X$, $R = A$, and π is the identity.

Suppose Player I plays v . If the image $w = \pi(v)$ is zero, Player II plays $\emptyset$ and leaves the state unchanged. Otherwise let $q : Q \rightarrow Q/\mathbb{R}w$ be the quotient map. By the standard Fubini theorem for Borel sets, the section-measure function

$$z \mapsto m_1(R \cap q^{-1}(z))$$

is measurable and

$$N = \{z \in Q/\mathbb{R}w : m_1(R \cap q^{-1}(z)) > 0\}$$

is a null Borel set in $Q/\mathbb{R}w$, because R is null in Q . Set

$$G = R \setminus q^{-1}(N).$$

Then every line in Q parallel to w meets G in a one-dimensional null set. Player II plays

$$B = \pi^{-1}(G).$$

This is a legal move: if L is any line in X parallel to v , then $\pi(L)$ is a line in Q parallel to w , and $L \cap B$ is identified with a subset of $\pi(L) \cap G$, hence has one-dimensional measure zero. Thus $B \in \mathcal{A}(v)$.

The uncovered part of A is now contained in

$$\pi^{-1}(R \setminus G) \subset \pi^{-1}(q^{-1}(N)).$$

Replace S by $S + \mathbb{R}v$, replace Q by $Q/\mathbb{R}w$, replace π by $q \circ \pi$, and replace R by N . The invariant is preserved and the dimension of the quotient has dropped by one.

A legal play by Player I must be dense in X , so its played directions span X . Therefore Player I eventually plays $\dim X$ linearly independent directions. After the last such move the quotient has dimension zero. In the one-dimensional final quotient step, the null set R has zero one-dimensional measure, so the positive-section set N is empty. Hence the remaining residual is empty. Thus A is covered by Player II's moves, and the described strategy wins. $\square$

Verification Notes

The proof uses only standard finite-dimensional Fubini disintegration for Borel sets under a linear quotient map. The moves played by Player II need not be subsets of A ; they only need to be Borel members of the corresponding $\mathcal{A}(v)$, as in the source game definition.

If one permits Player I to play the zero vector despite $\mathcal{A}(0)$ not being defined in the source paper, the strategy treats that inning as irrelevant and plays $\emptyset$. Equivalently, one may read the game as requiring non-zero directions, which is the only interpretation under which Player II's legal response is defined.

Limitations and Novelty Check

This packet proves only the finite-dimensional case. It does not address the source conjecture for infinite-dimensional separable Banach spaces, nor the other diagram-reversal questions in the paper.

Cheap run indexes had no previous packet for arXiv:0601556. Local corpus searches for “Aronszajn-null game” and exact phrases from the source conjecture found only the source paper. A bounded web search for the same phrases likewise found the source arXiv page and no later answer. Novelty confidence is moderate: the finite-dimensional argument is elementary and may be folklore, but it is not recorded in the run memory.

References

- J. Duda and B. Tsaban, *Games in Banach spaces*, arXiv:math/0601556.
- Y. Benyamini and J. Lindenstrauss, *Geometric Nonlinear Functional Analysis, Volume 1*, AMS Colloquium Publications 48, 2000.